

classmate

Activity

Stack Queue and IL GATE Questions

① $\text{push}(54), \text{push}(52), \text{pop}(), \text{push}(55), \text{push}(62),$
 $S = \text{pop}()$

Stack execution:

Push 54 $\rightarrow [54]$

Push 52 $\rightarrow [54, 52]$

Pop $\rightarrow$ removes 52

Push 55 $\rightarrow [54, 55]$

Push 62 $\rightarrow [54, 55, 62]$

$S = \text{Pop} \rightarrow S = 62$

Queue execution:

$[21]$

$[21, 24]$

Dequeue $\rightarrow$ removes 21

$[24, 28]$

$[24, 28, 32]$

$q = \text{Dequeue} = 24$

$$S + q = 62 + 24 = 86$$

② postfix expression

10 5 + 60 6 / * 8 -

$$105 + = 15$$

$$606 / = 10$$

$$15 \times 10 = 150$$

$$150 - 8 = 142$$

Ans: 142

3) The minimum no. of enqueue operations on Q1 required is 6

4) GATE CSE 2022 (Modified Stack with Reverse)
A language accepted by a pushdown automation with REVERSE operation is more powerful than a context-free language

5) GATE CSE 2014 Set 2

Postfix expression:
 $8\ 2\ 3\ ^{\wedge}/23\ *51\ * -$

Push 8 $\rightarrow [8]$

Push 2 $\rightarrow [8, 2]$

Push 3 $\rightarrow [8, 2, 3]$

$\wedge \rightarrow 2^3 = 8 \rightarrow [8, 8]$

$/ \rightarrow 8/8 = 1 \rightarrow [1]$

Push 2 $\rightarrow [1, 2]$

Push 3 $\rightarrow [1, 2, 3]$

First $* \rightarrow 2 * 3 = 6 \rightarrow [1, 6]$

Top two elements are 6 and 1 (top = 6)

6) GATE CSE 2007 (Infix $\rightarrow$ Postfix)
Infix:

$$a + b \times c - d \wedge e \wedge f$$

Since $\wedge$ is right associative:

$$\text{Postfix} = abc * + def \wedge \wedge - *$$

7) GATE CSF 2004

PUSH (10)

PUSH (20)

POP $\rightarrow$ 20

PUSH (10)

PUSH (20)

POP $\rightarrow$ (20)

POP $\rightarrow$ 10

POP $\rightarrow$ 10

PUSH (20)

POP $\rightarrow$ 20

Popped sequence is :

20, 20, 10, 10, 20